



Helios Moisture-Pickup Test — Results Report

Down insulation moisture absorption & desorption

Specimen	10.0 g Expedry Gold-treated duck down	Date	June 3, 2026
Apparatus	Helios moisture-pickup chamber (hanging capsule on load cell)	Prepared by	Andrew Peterson
Protocol	90 min: 60 min wet hold @ 80% RH, 30 min dry	Contact	andrew@fuze47.com

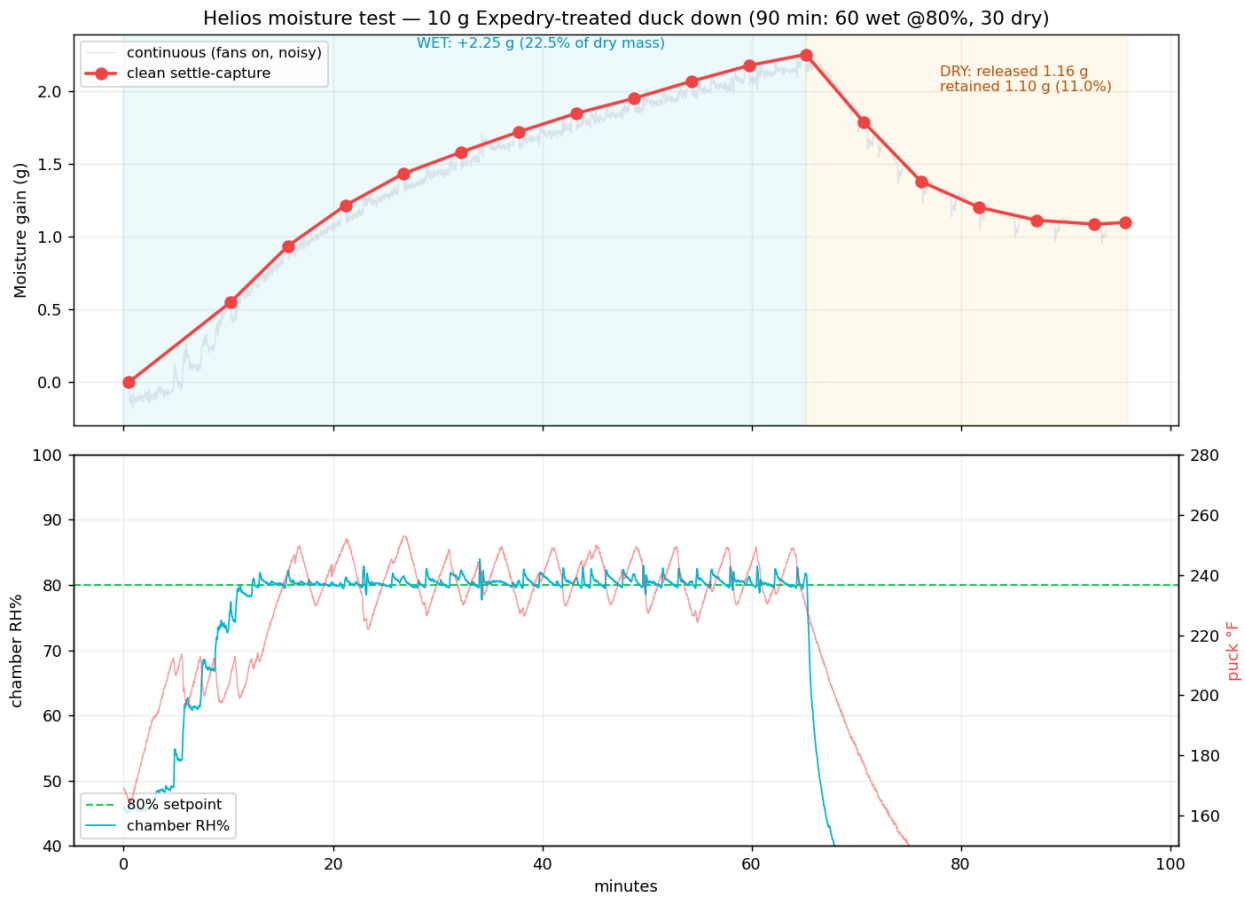
Summary

A 5.0 g specimen of Expedry Gold-treated duck down was held at 80% relative humidity for 60 minutes, then dried for 30 minutes by venting the chamber to 35% RH. Capsule weight was logged continuously, with airflow-free “settle-capture” weighings every 5 minutes (fans paused, capsule allowed to settle) providing the reported values. Over the wet phase the down absorbed **2.25 g of moisture (22.5% of its dry mass)**; the 30-minute dry phase released 1.16 g, leaving **1.10 g retained (11.0%)** at equilibrium with 35% RH air.

Results (airflow-free settle-capture weights)

Stage	Capsule weight	Moisture vs. dry baseline	% of dry mass
Dry baseline	0.00 g	—	—
End of 60-min wet hold	+2.25 g	+2.25 g absorbed	22.5%
End of 30-min dry	+1.10 g	released 1.16 g / retained 1.10 g	11.0% retained

Moisture curve



Red = airflow-free settle-capture weighings (reported values). Grey = continuous reading with fans running. Cyan = chamber RH against the 80% setpoint; red (lower) = puck temperature.

System stability during the 60-min wet hold

Channel	Mean	Std. dev.	Notes
Chamber RH	78.9%	5.3%	On target; SD inflated by 5-min settle windows pausing the drip loop
Puck temperature	234 °F	13 °F	Normal bang-bang control band
Chamber temperature	87.3 °F	2.3 °F	Stable
Differential pressure	-0.03 Pa	0.014 Pa	Essentially flat

Observations

Absorption was smooth and monotonic. The settle-capture curve shows a clean uptake with no scatter, and the continuous (fans-on) trace tracks it closely — confirming the airflow artifact is small and the reported weights are reliable.

The dry phase reached equilibrium. Retained mass flattened at roughly 1.10 g by ~87 minutes (about 25 minutes into the 30-minute dry), indicating the sample reached desorption equilibrium with 35% RH air. Extending the dry time at the same RH would not reduce it further; to drive moisture lower, target a drier chamber (lower dry-to-RH), not a longer hold.

Limitations & recommended next steps

- **Specimen mass corrected to 10 g.** The run was entered in the field as 5 g; the actual sample was 10 g. Absolute weights are unaffected (measured directly); all percent-of-dry-mass values in this report use the corrected 10 g basis.
- **Single run.** This is one complete characterization. Run-to-run repeatability requires 2–3 matched runs and is not yet established.
- **Control required.** The Expedry performance figure is the *difference* versus an untreated control run under the identical protocol; that comparison has not yet been performed.
- **Dry baseline.** For a quantitative absolute figure, the specimen should be desiccator- or oven-dried before the run; 45% uptake should be confirmed against a verified-dry starting mass.
- **Auto stabilize-and-tare** has since been added to the apparatus (the controller now waits for a stable reading and auto-zeroes before starting); this run pre-dates that feature and was tared manually.